Manpreet Singh

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Summary

ML Research Engineer with **5+** years of experience delivering DL and NLP solutions in production environments. Holds a **MSc in CS** with research focused on LLM interpretability and quantization. Skilled at taking machine learning projects from ideation to production, supported by successful delivery of multiple deployed systems at TCS.

Education

Master of Computer Science — Jan 2023 – Dec 2024

Dalhousie University – *Halifax, NS, CA*

Relevant coursework *– Machine Learning, Deep Learning, Natural Language Processing and ML for SE Applications.*

GPA (Graduate courses) – 3.9/4.3

Professional Experience

Research Assistant — May 2023 – Dec 2024

Hypermatrix Labs, Dalhousie University – *Halifax, NS, CA*

Interpreting the effects of Quantization on LLMs

- Conducted first of kind research by investigating multiple LLMs across 3 **quantization** settings, analyzing its impact on model behavior and neurons learning ability.
- Provided insights to improve NLP models by **identifying redundant and dead neurons.**
- Utilized **Integrated Gradients** to identify over **100 salient neurons** influencing model predictions out of **10,000+**, assessing model sensitivity to input features.
- Developed an **automated pipeline** using open-source LLMs to generate and validate neuron behavior in natural language, reducing explanation validation costs by 10 percent.
- **Published findings** in a thesis, contributing to the growing body of research on LLM optimization and explainability, with a **conference publication currently in progress.**

Manifold Learning in NLP Embeddings

- Evaluated **5+ manifold learning techniques** (Isomap, LLE, Hessian Eigenmapping, Spectral Embedding, t-SNE etc.) for dimensionality reduction of NLP token embeddings.
- **Visualized and analyzed** embedding spaces to identify **model-learned biases** within high-dimensional embeddings.
- Delivered actionable insights on algorithm performance, enabling more equitable and unbiased NLP model training pipelines.

AI Engineer — Jul 2019 – Oct 2022

iGnite Labs, Tata Consultancy Services – *Chennai, IN*

- Developed **Natural Language to SQL executer** app using GPT-3 for **custom DBs**, processed **500+** daily queries with **92%** accuracy.
- Deployed **Rasa-powered** NLP chatbot for increasing developer productivity handling **1000+** daily developer queries, improving productivity by **25%**.
- Created Computer Vision pipeline converting **2D blueprints to 3D models**, reducing visualization time by **20%**.

- Automated **15+** dev tasks using Python and shell scripting, saved **20+** hours weekly.
- Mentored **100+** engineers in ML tools and best practices; **95%** training completion rate.

Publications

- Manpreet Singh, & Hassan Sajjad. (2025). Interpreting the Effects of Quantization on LLMs. [https://arxiv.org/abs/2508.16785]

Key Skills

- **Programming Languages:** Python (primary), Java, JavaScript, C
- **Machine Learning & AI:** PyTorch, TensorFlow, Production RAG, Agentic-AI, Prompt Engineering, Rasa platform, Explainable AI, Model Interpretability, Transfer Learning, Contrastive learning, RLHF, Finetuning, Quantization, Hyperparameter tuning
- **Data Analysis:** SQL, EDA, Feature Engineering, Experiment design, Model Evaluation
- **High-Performance & Distributed Computing:** Slurm management, Multicore Optimization, Cluster Management
- **Software Development & DevOps:** Docker, Kubernetes, Git, Flask, Django, RESTful APIs
- **Management & Leadership:** Agile Methodologies, Technical Project Management, Team Leadership, Mentorship

Certifications

- Academy Accreditation - Databricks Fundamentals [Databricks] – Data ingestion, processing, and collaborative analytics
- LLMOps [DeepLearning.AI] – End-to-end ML pipelines with BigQuery, Vertex AI, and Kubeflow
- Getting Started with Deep Learning [Nvidia] – Hands-on with data augmentation, model training, and evaluation

Extracurricular Achievements

- Presented Neural Style Transfer project to India’s Education Minister (TCS internal showcase).
- Ranked in top 3 of 1000 participants in a 20-week AI/ML training program at TCS.

References

Available on request.